



T.J.I.A. JAYAWICKRAMA

BSc. (Hon's) in Computer Engineering (UG)

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 [Induwara Jayawickrama](#)

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PROFILE

I am an enthusiastic, self-motivated and reliable undergraduate student with a strong passion for Cyber Security, IoT Systems, Software Engineering and Embedded systems. I thrive in both collaborative team environments and when working independently. I am highly adaptable to new and challenging situations and I approach problems with a proactive mindset and a keen attention to detail. I perform well under pressure, consistently meeting tight deadlines while maintaining quality and precision. My goal is to contribute meaningfully to innovative projects while continuing to learn and grow in the industry.

EDUCATIONAL QUALIFICATIONS

Bachelor of Science in Computer Engineering
University of Jaffna, Sri Lanka

2022 to Present

Relevant Coursework: Embedded Systems Design, Computer Architecture, Robotics and Automation, Control Systems, Software Engineering, Object-Oriented Programming, Database Management Systems, Computer Networks & Security, Operating Systems, Data Structures & Algorithms, Machine Learning, Deep Learning, Artificial Intelligence, Advanced Networking, Operational Research.

PROJECTS

Bookify – Service Booking Web Application (Group Project) 

01/2025 – 04/2025

Spring Boot | Reactjs | MySQL | RESTful APIs

Bookify is a full stack service booking web application developed as a group project. It connects consumers with service providers, allowing users to browse services, view provider profiles, check availability and make bookings. Service providers can register on the platform, manage their profiles and receive bookings.

Face2Face – Real-Time Video Chat Application 

02/2025 – 04/2025

WebRTC | Socket.IO | JavaScript | HTML | CSS | Node.js | Express.js | Railway | Firebase | Vercel | GitHub

Developed a browser based peer to peer video chat application using WebRTC and Socket.IO. The platform enables users to join rooms, share video/audio streams, exchange text messages and toggle screen sharing and recording functionality. The project emphasized real time communication, media stream handling and frontend backend integration without relying on third party video services

Accident Detection and Prevention System 

02/2025 – 04/2025

ESP32 | Ultrasonic Sensor | GSM | GPS | Flame Sensor | C++ | Node.js | Google Maps API | GitHub

Developed a smart embedded system that detects potential road accidents and initiates emergency communication. Utilized ultrasonic sensors and ESP32 for real time obstacle and pedestrian detection. Integrated GSM and GPS modules to send automated alerts with location data to emergency services. Focused on pre accident prevention and post accident response, ensuring driver awareness and timely intervention through a web app interface.

Blockchain-Enhanced Security for Lightweight IoT Embedded Systems (Research)

01/2025 – Present

IoT | Blockchain | ECDSA | AES-128 | SHA-256 | MQTT | Python | Mininet | SQLite | Wireshark

Researching a lightweight block chain framework to secure IoT systems in resource constrained environments. Designed a three layer architecture using AES-128 for encryption, ECDSA for authentication and a simplified Proof of Stake consensus. Implemented secure communication via MQTT and developed validator logic in Python. Evaluated system performance through real hardware testing and simulation using Mininet and Wireshark, focusing on metrics such as energy consumption, latency, and attack resilience across heterogeneous IoT scenarios.

TECHNICAL SKILLS

Programming Languages | Java, C++, C, Python, PHP, HTML, CSS, JavaScript, MATLAB
Frameworks | React, Express.js, Spring Boot
Databases | MySQL, MongoDB
Tools & Platforms | Google Colab, Jupyter Notebook, Git, Postman, Vercel, Firebase, Railway
Embedded Systems | Arduino, Embedded C
Network and Security | Wireshark, WebRTC, Cisco Packet Tracer

SOFT SKILLS

- Problem Solving
 - Communication Skills
- Critical Thinking
 - Teamwork
- Flexibility
 - Creativity
- Leadership
 - Adaptability

CERTIFICATES

- Lead Yourself Certificate of Completion Personality Development Program
- Time Management (From Chegg)
- Certified dark web analyst (Dark web and Deep web course)
- YarInsight 2.0 - Summer School on Full Stack Web Development with the MERN Stack

VOLUNTEERING

Web Master of IEEE PES Student Chapter - University of Jaffna

04/2024 – present

Managed and maintained the Chapter's website, organized virtual events via Zoom and Google Meet and created Google Forms for registrations, feedback, etc. Ensured seamless online communication and timely updates of Chapter activities. Supported the team with technical aspects of digital outreach.

Sub Editor of Engineering Student Union- University of Jaffna

08/2023 – 08/2024

Created and curated engaging posts for various events, ensuring clear communication and promotion of Union activities across digital platforms. Collaborated with the team to maintain consistent content quality and timeliness.

Sub Web Master of IEEE PES Student Chapter - University of Jaffna

04/2023 – 04/2024

Assisted in maintaining the Chapter’s website and ensured regular updates. Supported the management of Zoom meetings for events and sessions.

COMPETITIONS

IEEEXtreme 18.0	108 th all island and 6 th place of SBC UOJ	2024
CodeRally 5.0	Rank 12	2024
SLRC line following robotics competition		2024
MoraXtreme		2023, 2024

PROFESSIONAL MEMBERSHIPS

Student Member: Institution of Engineers Sri Lanka (IESL), Membership ID: S-29538	2023 - present
Student Member: IEEE Sri Lanka Section	2023 – present

REFERENCES

Dr. (Eng.) A. Kaneswaran	Dr. (Mrs.) J. Jananie
Dean,	Senior Lecturer,
Senior Lecturer Grade II,	Department of Computer Engineering,
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